KHULNA UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Courtesy:
Dr. Md. Selim Hossain

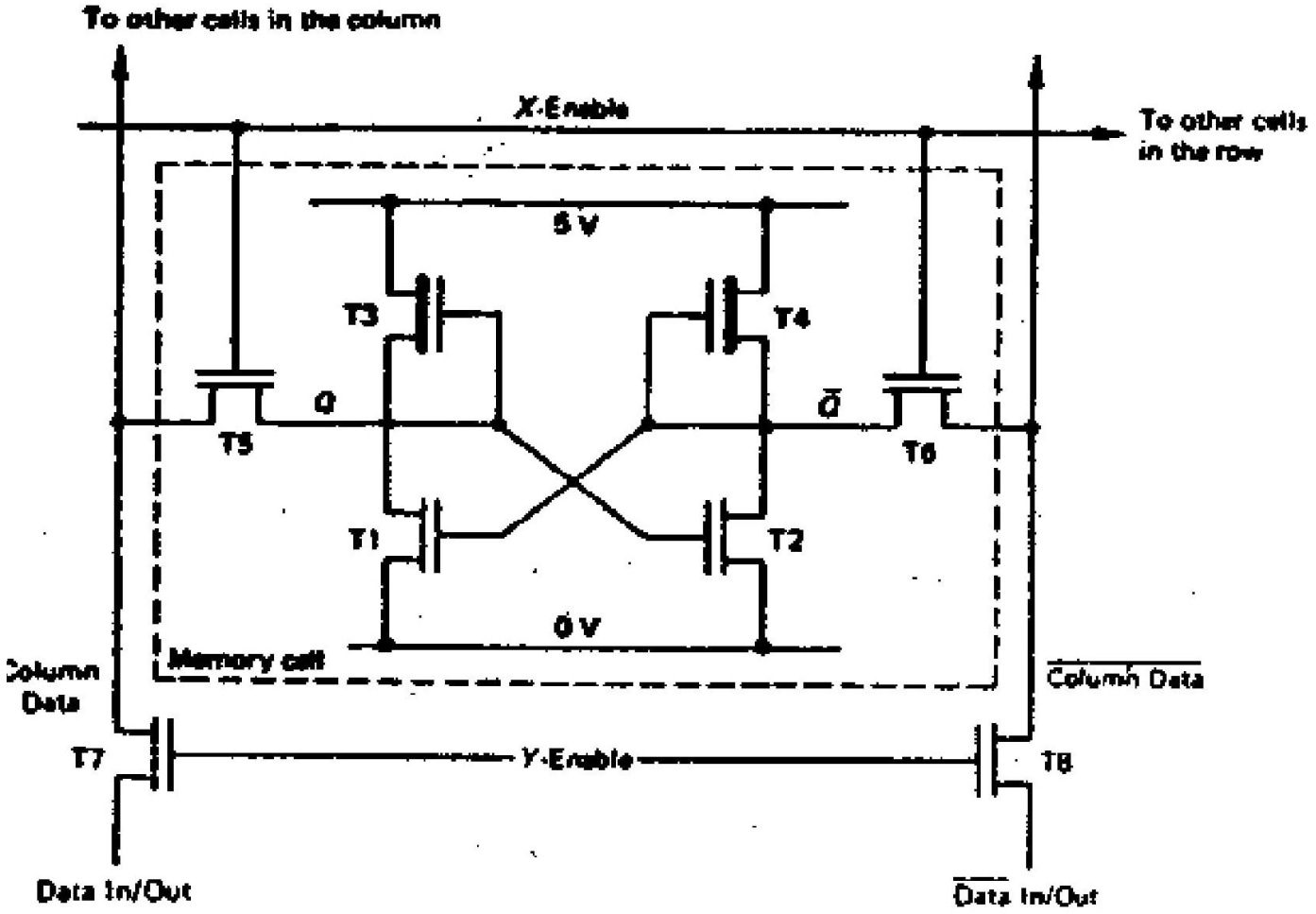
Department of Electrical and Electronic Engineering

Course Code : EE 4121

VLSI

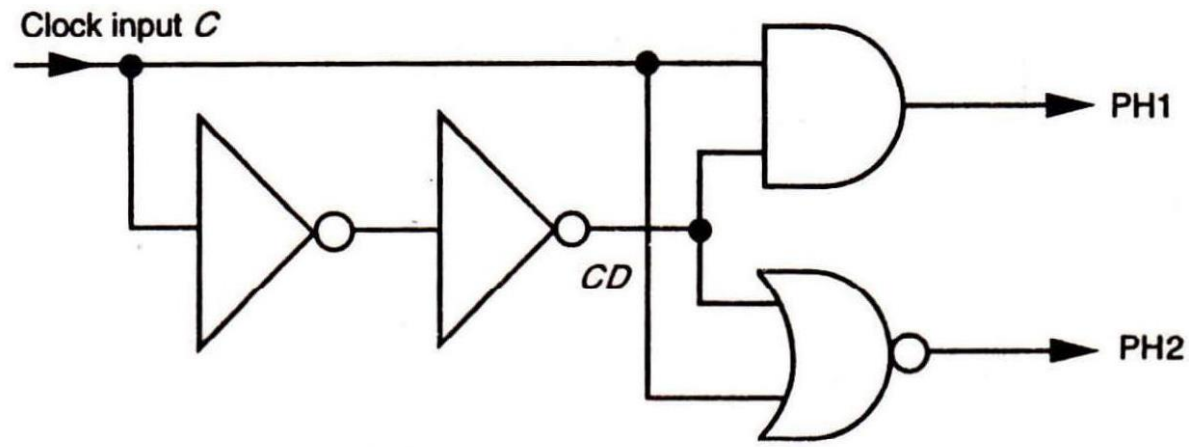
Six-Transistor Static Memory Cell

- It consists of two cross-coupled inverters which form an FF, plus two pass Transistors T5 and T6.
- If Reading, the X-enable signal is applied turning on T5 and T6. Thus, the Q and Q-bar FF outputs appear on Column Data and Column Data-bar lines, respectively.
- The Y-enable or the selected column turns on T7 and T8, so that the data can be clocked into the output reg.
- Read/Write are not executed at the same time.
- If read is execute in the first phase, then write will be executed in the next phase.
- Here six transistors, so it will be more expensive.



Two-phase Clock Generator Circuits

- A two-phase clock offers a great deal of freedom in sequential circuit design if the clock period and duration of the signal PH1 and PH2 are correctly chosen.
- Race-around condition, timing conflict arises if two-phase clocks are not used.
- The input clock signal C is used to provide a delayed version of itself (CD) by passing it through an even no. Of inverters.
- PH1 and PH2 can never be on simultaneously.
- The inverters are used for creating a time-delay.
- More inverters can be added to increase the delay.



URE 6.33(a) Simple two-phase clock generator circuit—basic form.

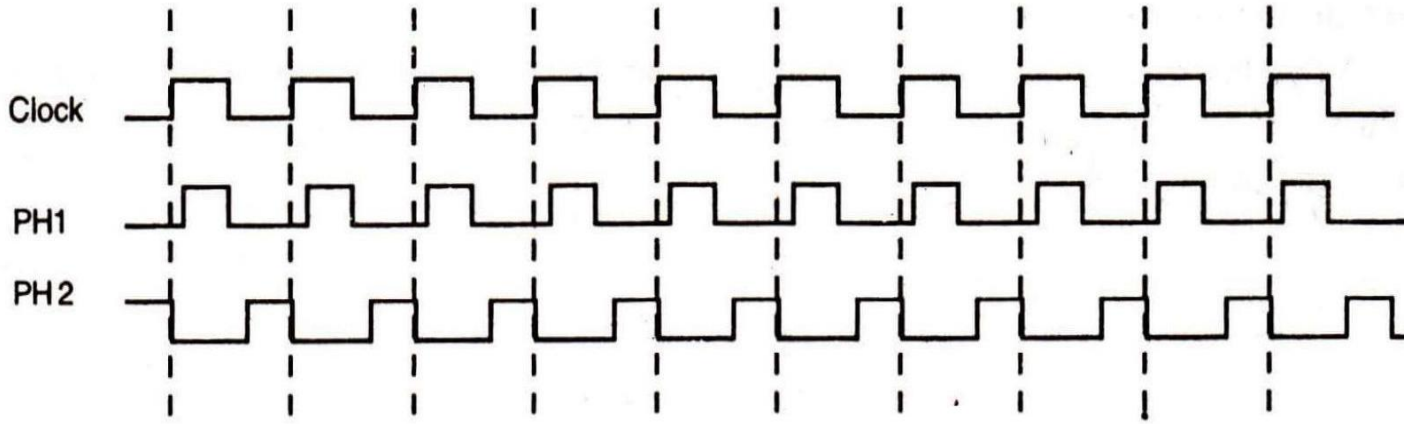
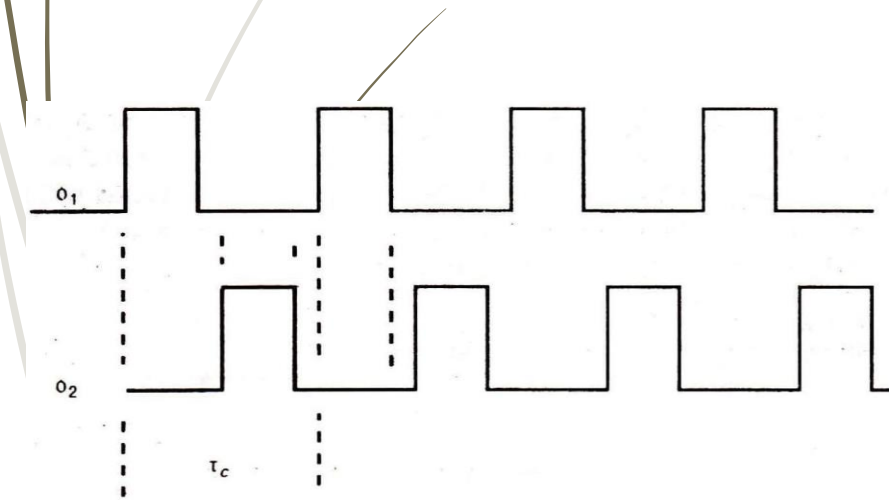


FIGURE 6.33(b) Waveforms for two-phase clock generator.

Two-phase Clock Generator Circuit: Continue

- When studying Figure 6.31, it is necessary to recognize the fact that φ1 and φ2 do not need to be symmetrical as shown. For a given clock period, each clock phase period and its associated underlap period can be divided if the need arises in optimizing a design.
- A number of techniques are used to generate the two clock phases. One popular method is illustrated in Figure 6.32 and it will be seen that the output frequency is one-quarter of that of the input clock.



Notes: 1. τ_c = clock period
2. $o_1(t)$, $o_2(t) = 0$; *all* t

FIGURE 6.31 Two-phase clocking.

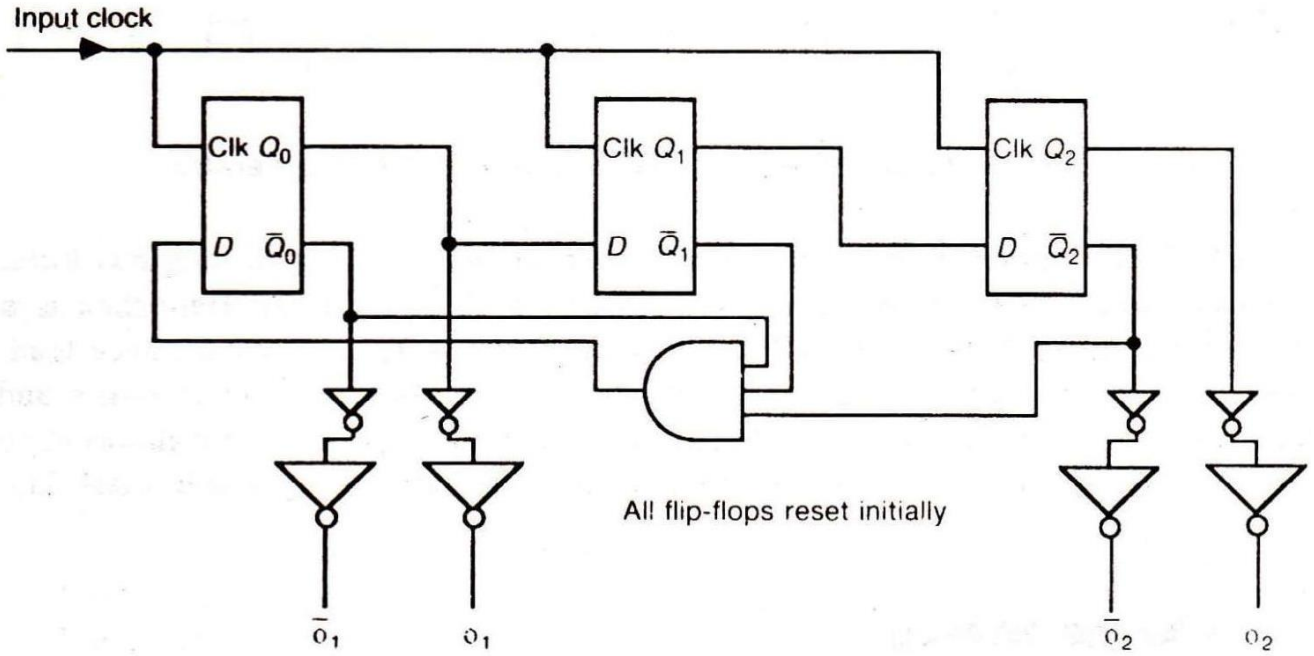


FIGURE 6.32 Two-phase clock generator using D flip-flops.

System Timing Considerations

1 SYSTEM TIMING CONSIDERATIONS

1. A two-phase non-overlapping clock signal is assumed to be available, and this clock alone will be used throughout the system.
2. Clock phases are to be identified as ϕ_1 and ϕ_2 where ϕ_1 is assumed to lead ϕ_2.

3. Bits (or data) to be stored are *written to* registers, storage elements, and subsystems on ϕ_1 of the clock; that is, write signals *WR* are *Anded* with ϕ_1.
4. Bits or data written into storage elements may be assumed to have settled before the immediately following ϕ_2 signal, and ϕ_2 signals may be used to *refresh* stored data where appropriate.

System Timing Considerations: Continue

5. In general, delays through data paths, combinational logic, etc. are assumed to be less than the interval between the leading edge of ϕ_1 of the clock and the leading edge of the following ϕ_2 signal.
6. Bits or data may be *read* from storage elements on the next ϕ_1 of the clock; that is, read signals *RD* are *Anded* with ϕ_1. Obviously, *RD* and *WR* are generally mutually exclusive to any one storage element.
7. A general requirement for system stability is that there must be at least one clocked storage element in series with every closed loop signal path.

Strict adherence to a set of rules such as this will greatly simplify the task of the system designer and also help to avoid some of the disasters which will almost certainly occur if a haphazard approach is taken.

Comparative Assessment of Storage Elements

In order to make a comparative assessment of some possible storage elements, we will consider the following factors:

- area requirement;
- estimated dissipation per bit stored;
- volatility.

9.2.1 The Dynamic Shift Register Stage

One method of storing a single bit is to use the shift register approach previously introduced in section 6.5.4 (and also Figures 3.14, 3.17, 6.36, 6.37, 6.38, 6.39 and 6.40).

Comparative Assessment of Storage Elements: Continue

9.2.1.1 Area

This calculation applies to an nMOS design, as in Figure 6.40(a), with buried contacts. Allowing for the sharing of V_{DD} and *GND* rails between adjacent rows of register cells, each bit stored will require

$$(22\lambda \times 28\lambda) \times 2 \doteqdot 1200\lambda^2$$

For example, for $\lambda = 2.5\ \mu m$

$$\text{Area/bit} = 7500\ \mu m^2$$

Comparative Assessment of Storage Elements: Continue

To give an idea of what this implies, such area requirements would result in a maximum number of bits stored on a 4 mm × 4 mm chip area ≈ 2.1 k bits.

For a CMOS design, as in Figure 6.40(b), and allowing for the sharing of V_{DD} and V_{SS} rails between adjacent rows of register cells, each bit stored will require

$$(38\lambda \times 28\lambda) \times 2 \doteqdot 2100\lambda^2$$

For example, for $\lambda = 2.5\ \mu m$

$$\text{Area/bit} \approx 13{,}000\ \mu m^2$$

Such area requirements would result in a maximum number of bits stored on a 4 mm × 4 mm chip area ≈ 1.2 k bits.

Comparative Assessment of Storage Elements: Power Dissipation

Dynamic Power Dissipation:

In the case of CMOS designs, the static dissipation is very small and calculation at this stage will not be meaningful since only the switching dissipation will be significant (particularly at high speeds). This dynamic power consumption P_d can be written as

$$P_d = m.(C_L.V_{DD}^2.f)$$

where m is the duty cycle, C_L is effective load capacitance and f is the clock frequency.

Comparative Assessment of Storage Elements: Power Dissipation….Continue

Static Power Dissipation:

In the nMOS case we can readily calculate the static dissipation, noting that in practice the switching dissipation would add to this. Each inverter stage has a ratio of 8:1 and if the layout of Figure 6.40(a) (buried contacts) is used, then, noting that one inverter of the pair must always be 'on',

$$Z_{p.u.} = 4R_s$$

and

$$Z_{p.d.} = \frac{1}{2}R_s$$

Therefore

$$\text{Current} = \frac{V_{DD}}{Z_{p.u.} + Z_{p.d.}} = \frac{5\,\text{V} \times 10^6}{4.5 \times 10^4} = \frac{500}{4.5}\,\mu\text{A} \approx 110\,\mu\text{A}$$

Comparative Assessment of Storage Elements: Power Dissipation….Continue

Static Power Dissipation..Continue:

Therefore

$$\frac{\text{Static dissipation}}{\text{Bit stored}} = V_{DD} \times \text{current} = 5\ \text{V} \times 110\ \mu\text{A} = 550\ \mu\text{W}$$

Thus, 2.1 k bits on a single chip would dissipate

$$2.1 \times 10^3 \times 550 \times 10^{-6} = 1.15\ \text{watts}$$

Dissipation can be reduced by using alternative geometry, but this is at the expense of increased area.

Comparative Assessment of Storage Elements: Volatility

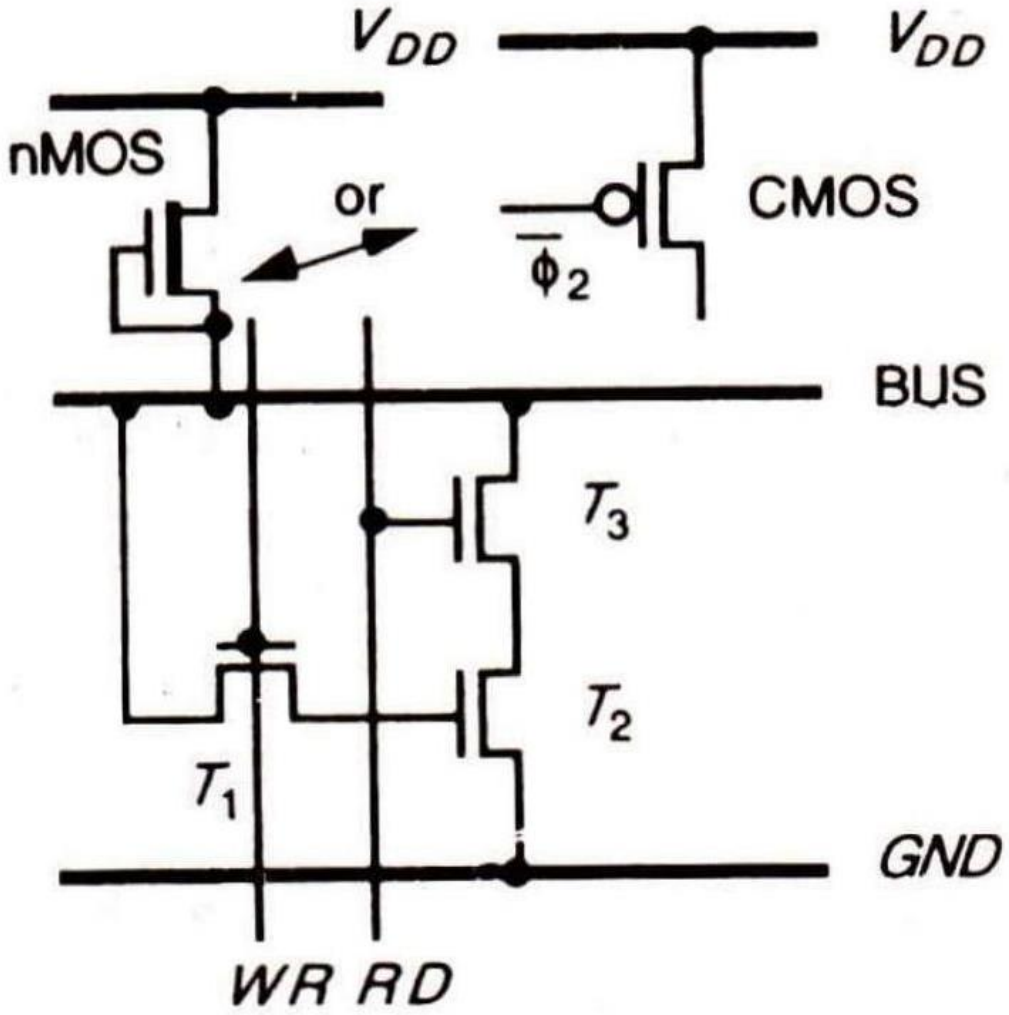
9.2.1.3 Volatility

Data is stored by the charge on the gate capacitance of each inverter stage, so that data storage time (without refresh) is limited to 1 msec or less.

Static Memory Cell: Non-volatile

Dynamic Memory Cell: Volatile

Three Transistor Dynamic RAM Cell

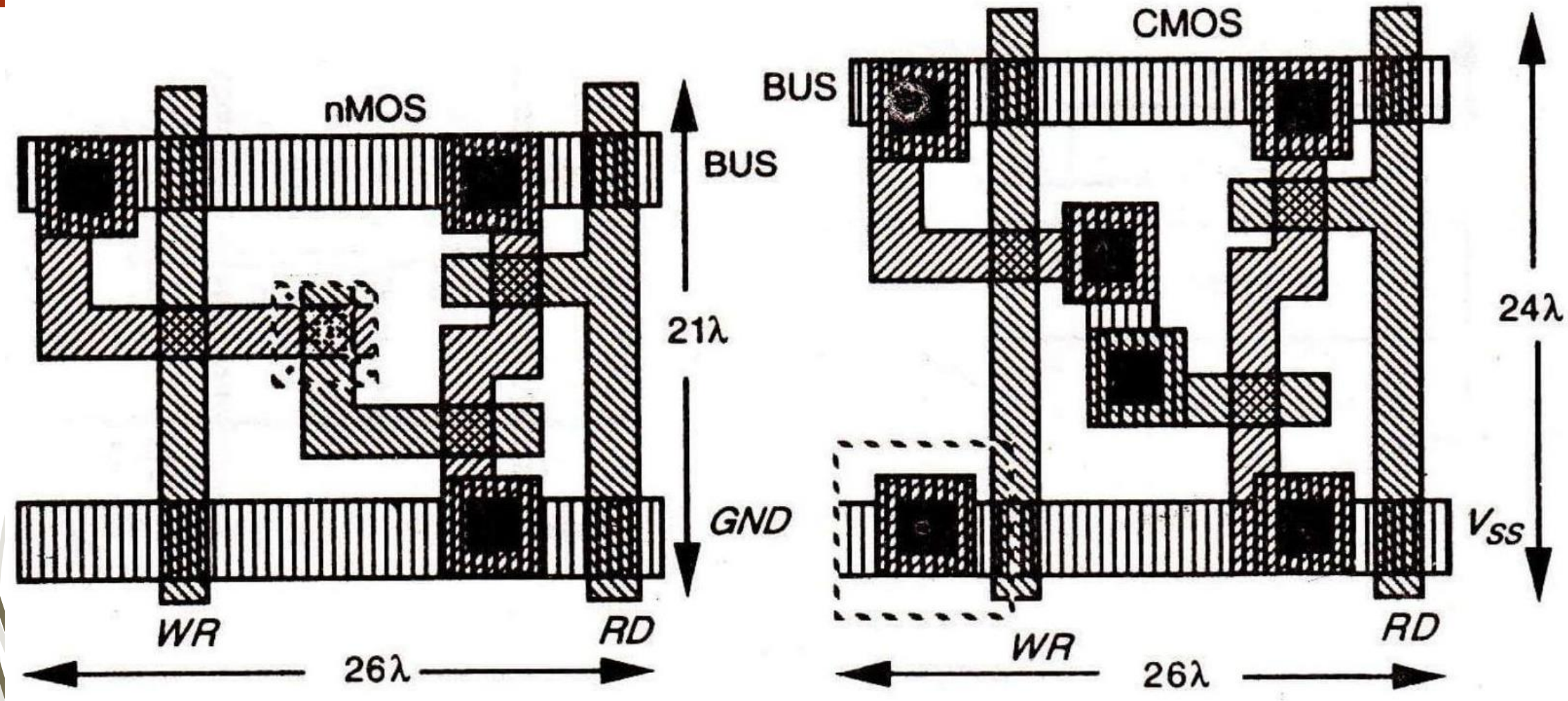


(a) Circuit

Note: *WR* and *RD* are coincident with ϕ_1.

- When WR is high, then RD is low
- Data of Bus stored in Cg of T2
- When both WR and RD are low data will remain in Cg of T2 till discharge.
- When RD is high and WR is low, store value high-> Bus will be low and store value low-> bus will be high which mean complement, Bus read the stored data in complement form

Three Transistor Dynamic RAM Cell: Continue



Three Transistor Dynamic RAM Cell: Continue

9.2.2.1 Area

From the layout it will be seen that an area of more than $500\lambda^2$ is required for each bit stored (less if *GND* (V_{SS}), and/or bus, and/or control lines are shared with other cells). Thus, for $\lambda = 2.5$ μm.

$$\text{Area/bit} \doteqdot 3000\ \mu m^2$$

Thus, to continue the previous example, the maximum number of bits which could be accommodated on a 4 mm × 4 mm silicon chip is > 4.8 k bits.

Three Transistor Dynamic RAM Cell: Continue

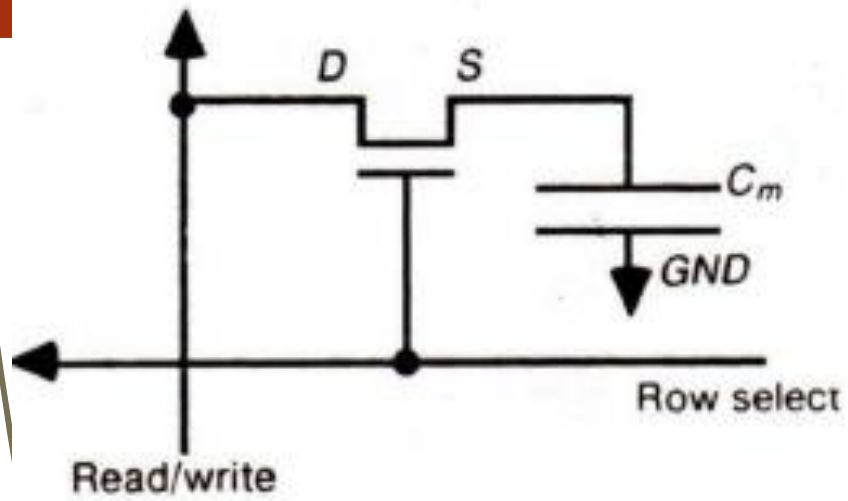
9.2.2.2 Dissipation

Static dissipation is nil since current flows only when *RD* is Hi and a 1 is stored. Thus, the actual dissipation associated with each bit stored will depend on the bus pull-up and on the duration of the *RD* signal and on the switching frequency.

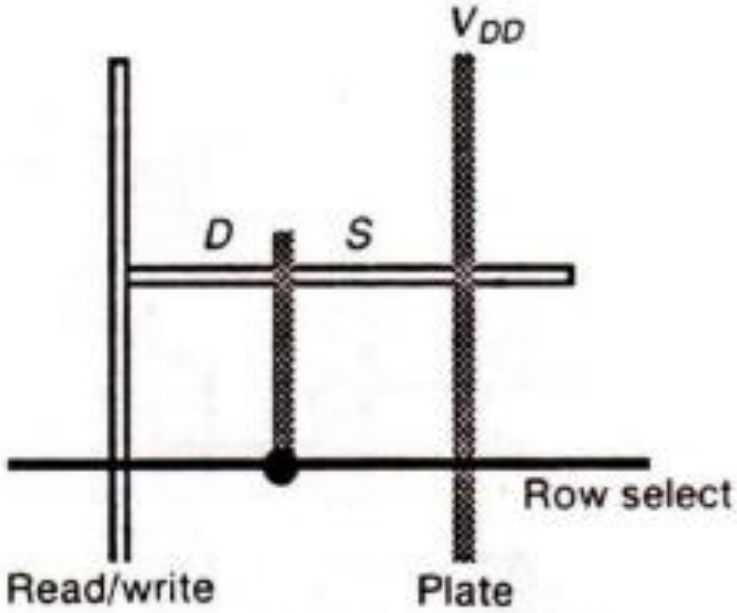
9.2.2.3 Volatility

The cell is 'dynamic' and will hold data only for as long as sufficient charge remains on C_g (of T_2).

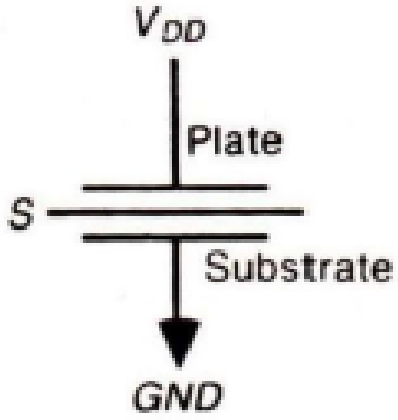
One Transistor Dynamic RAM Cell



a) Circuit arrangement



(b) Stick diagram



(c) Equivalent Circuit

- Here, we need to use one direct capacitor Cm.
- RD, WR will not be performed simultaneously.
- Cm to be charged when row select high and WR is high from WR/RD select line.
- Cm-> very difficult to distinguish 1& 0, so sense amplifier can be used to sense the output but complex.
- Cm-> need sufficient value or large capacitor is desired, but area can be increased,
- So we need special type of capacitor whose value should be 0.125 pf.

One Transistor Dynamic RAM Cell: Continue

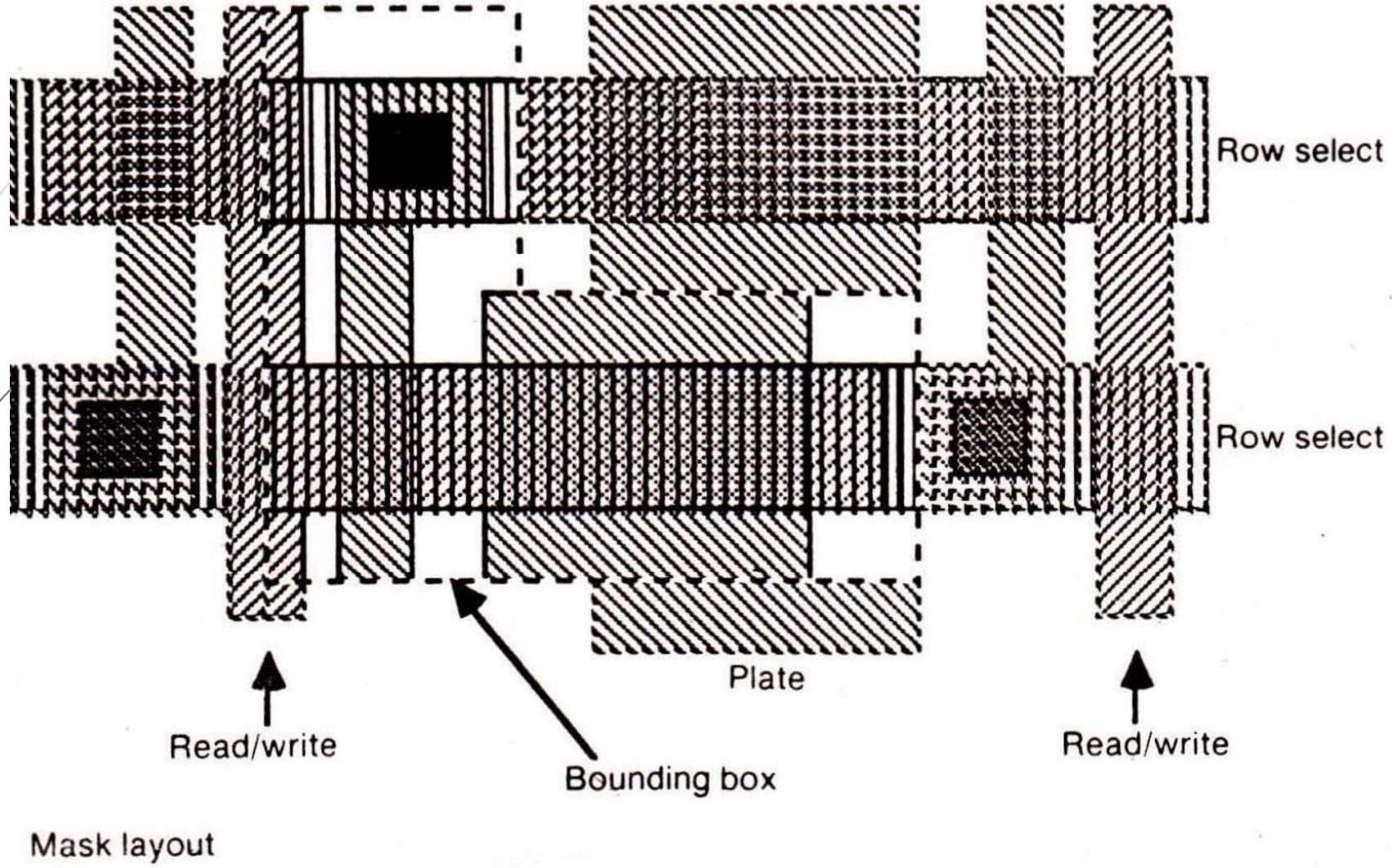


FIGURE 9.3 One-transistor memory cell.

One Transistor Dynamic RAM Cell: Continue

9.2.3.1 Area

The area enclosed to indicate the standard cell in Figure 9.3(d) is $200\lambda^2$. Thus for $\lambda = 2.5\ \mu m$, area/bit stored $= 200\lambda^2 = 1250\ \mu m^2$.

Therefore, the number of bits per 4 mm × 4 mm chip area is approximately 12 k bits (allowing some 'overheads' for sensing, etc.).

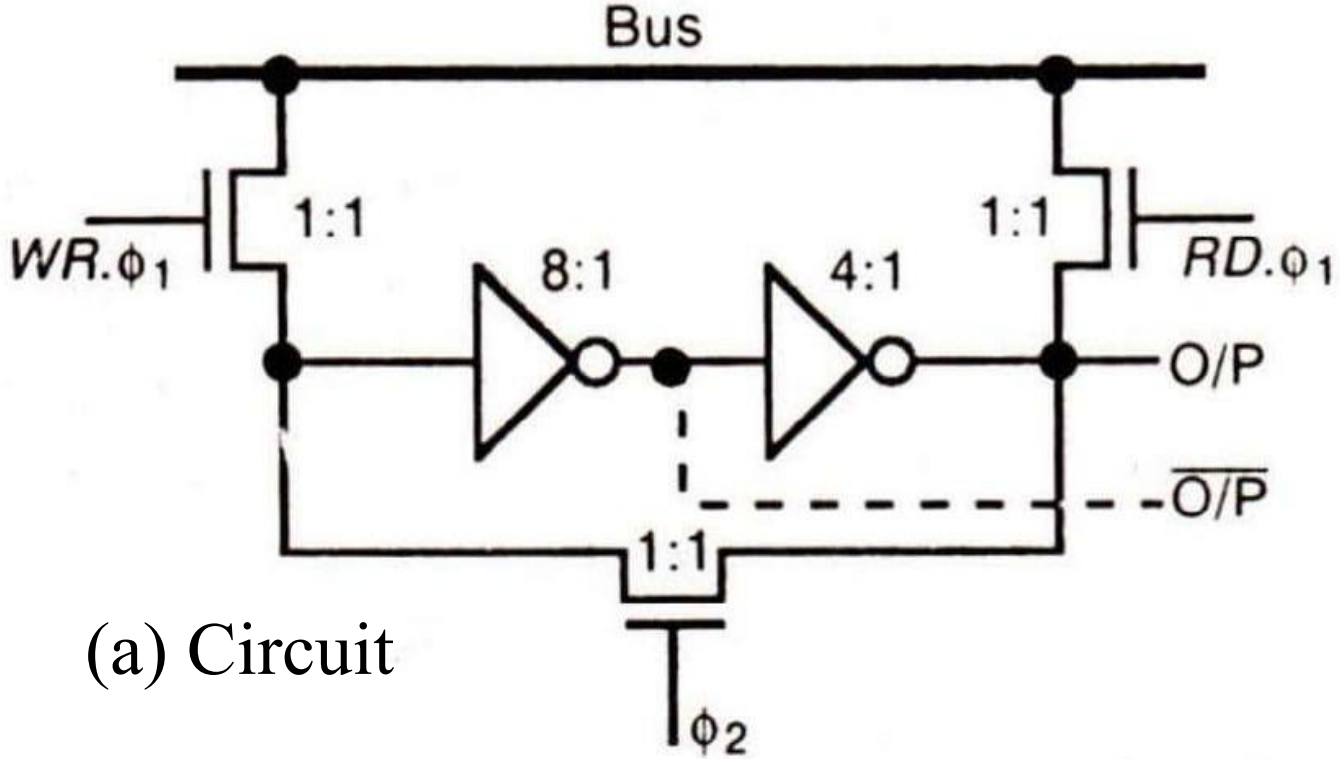
9.2.3.2 Dissipation

There is no static power associated with the cell itself, but there must be an allowance for switching energy while writing to and reading from the storage elements.

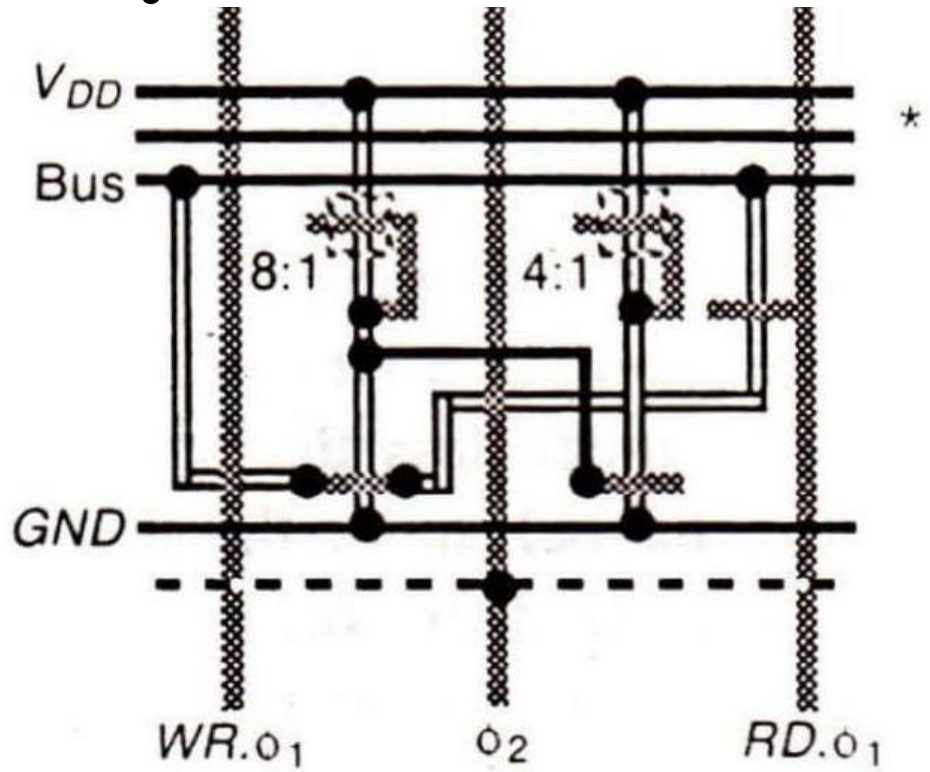
9.2.3.3 Volatility

Quite obviously, leakage current mechanisms will deplete the charge stored in C_m and thus the data will be held for only up to 1 msec or less. Therefore, periodic refresh operations must be provided. It will also be realized that reading the cell is a destructive operation and that the stored bit must be rewritten every time it is read.

nMOS Pseudo-Static Memory Cell



(a) Circuit



(b) Stick diagram

- **When on RD then next will be WR, So WD and WR are mutually exclusive.**
- **Φ2 will be refreshed in each clock pulse**
- **Φ2 of the clock occurs at the interval less than the decay time of the stored bit**

FIGURE 9.4 nMOS pseudo-static memory cell.

nMOS Pseudo-Static Memory Cell: Continue

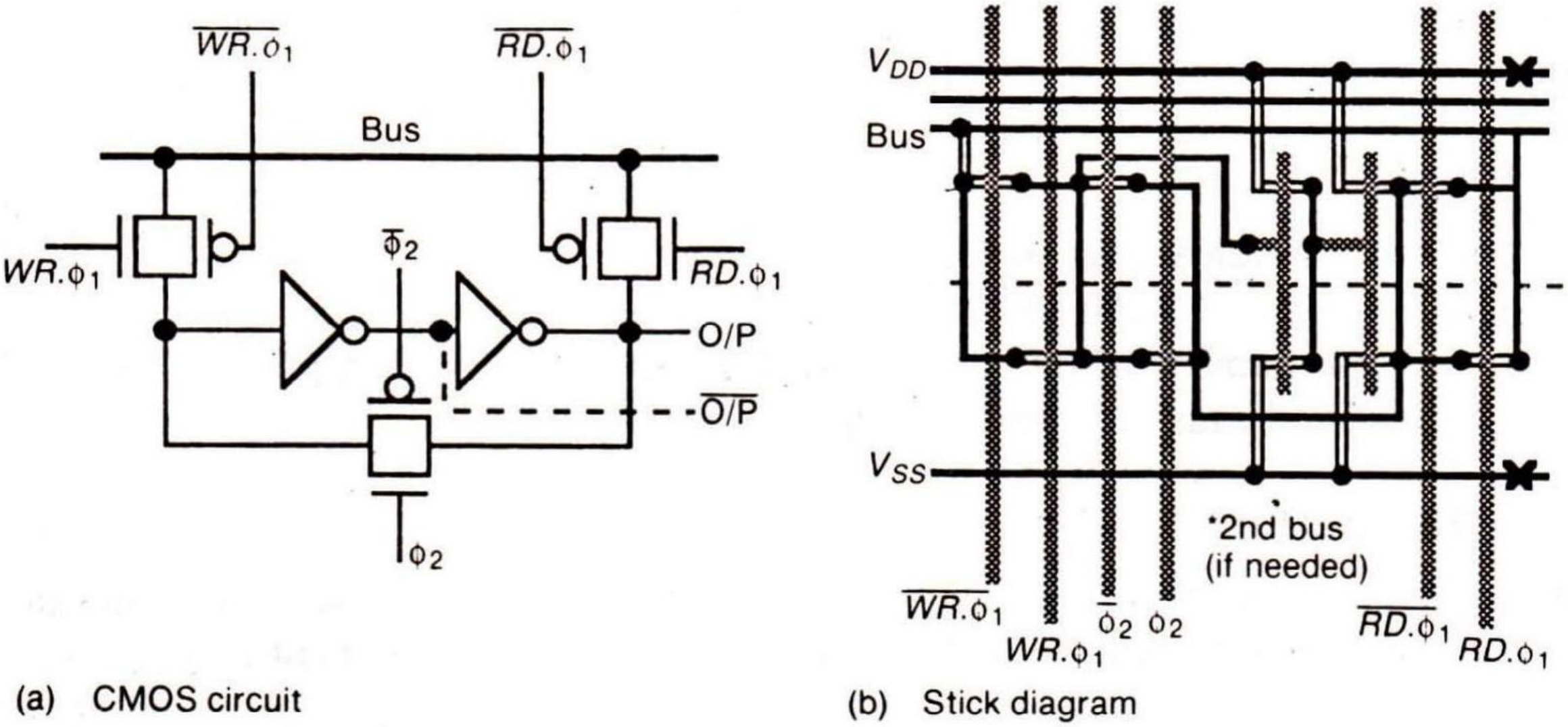


FIGURE 9.5 CMOS pseudo-static memory cell.

nMOS Pseudo-Static Memory Cell: Continue

9.2.4.1 Area

A typical area for a nMOS single cell with single bus is in the region of $1750\lambda^2$. Therefore, for $\lambda = 2.5\ \mu m$

$$\text{Area/bit} \approx 10000\ \mu m^2$$

Thus, the maximum number of bits of storage per 4 mm × 4 mm chip is approximately 1.4 k bits.

nMOS Pseudo-Static Memory Cell: Continue

9.2.4.2 Dissipation

The nMOS cell uses two inverters, one with an 8:1 and the other with a 4:1 ratio. Dissipation will depend on the current drawn and thus on the actual geometry of the inverters, but let us assume that inverters are based on minimum feature size gate areas so that the 8:1 stage will present a resistance of 90 kΩ and the 4:1 stage a resistance of 50 kΩ between the supply rails. Now when one stage is off, the other is on so that, say, each spends half-time in the conducting state. Therefore

$$\text{Average current} = 0.5\left(\frac{5\text{ V}}{90\text{ k}\Omega} + \frac{5\text{ V}}{50\text{ k}\Omega}\right) \approx 80\ \mu\text{A}$$

Therefore dissipation per bit stored = 80 μA × 5 V = 400 μW. Thus 1.4 kbits on a single chip would dissipate 560 mW.

9.2.4.3 Volatility

The cell is non-volatile provided that ϕ_2 signals are present.